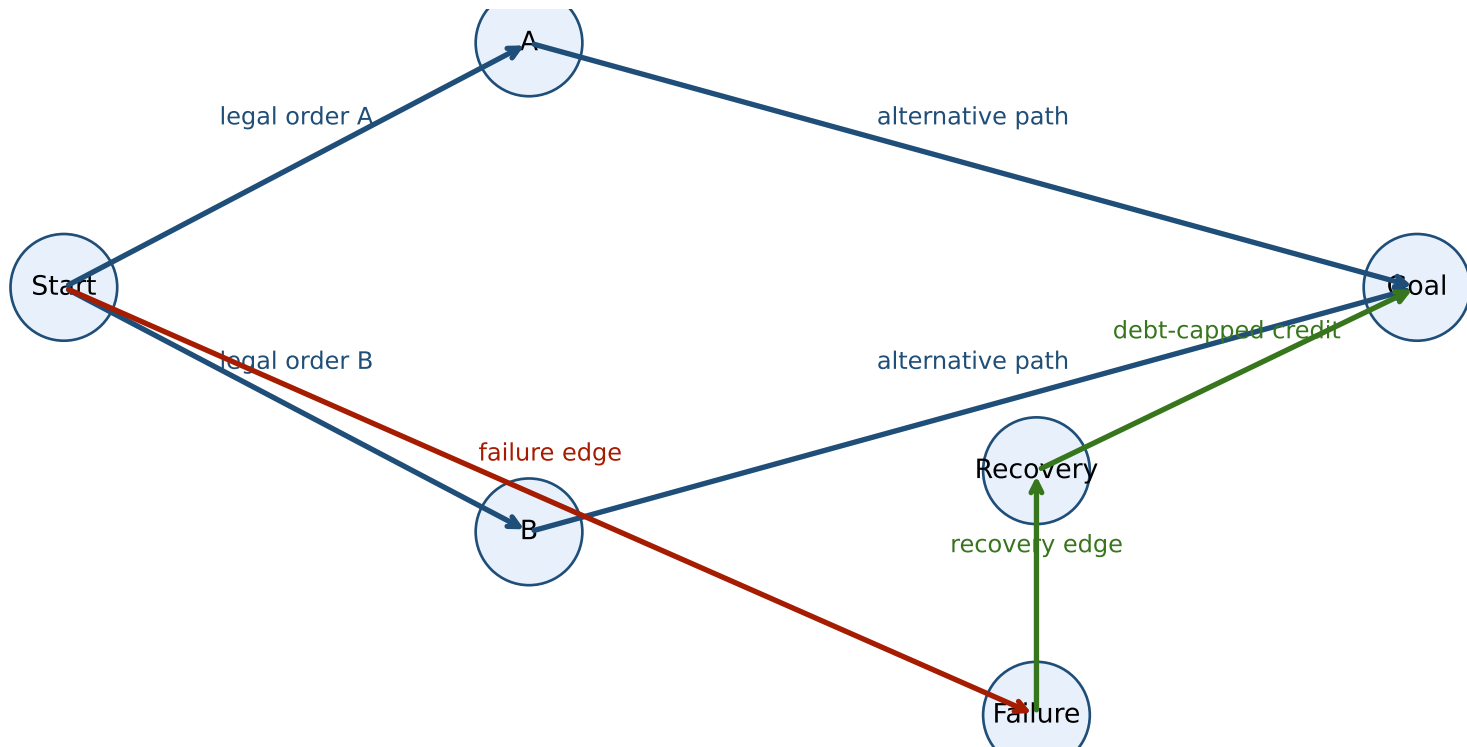


Locked manual PathGraph reward representation

$r = \Delta \text{ remaining-cost} + \lambda \Delta \text{ within-node progress} - \eta \text{ loop count}$
locked $\eta=0.0$, $\beta=0.0$; recovery credit is capped by failure debt



RA-BC
secondary downstream use